

Push-buttons & Buzzers

DURATION 45–50 Minutes	KEY CONCEPTS Inputs & Outputs, Complete Circuits, Hardware Control
MATERIALS NEEDED Microcontroller (e.g., micro:bit/Arduino), Push-button module, Active Buzzer module, 4x Jumper wires	LEARNING OBJECTIVES Understand the difference between an input device (button) and an output device (buzzer) by building a working physical circuit.

1. Introduction & Core Concepts (10 Mins)

Every electronic gadget we use has two main types of parts that let it interact with the world: **Inputs** and **Outputs**.

- **Input (The Sensor/Trigger):** Sends signals *into* our computer system. A push-button is a perfect example of an input hardware device. When you press it, it tells the computer, "Hey, someone is interacting with me!"
- **Output (The Actor/Responder):** Receives instructions *from* our computer system to make something happen in the real world. A buzzer is an output hardware device that converts electricity into sound energy.

Fun Concept: The Circuit Equation

Think of electricity flowing through your circuit like water through a pipe. We can describe a working connection using simple logic: **Signal_{Input} = 1 $\rightarrow$ Action_{Output} = ON**. When the circuit closes, the action happens immediately!

2. Hardware Blueprint & Setup (15 Mins)

To make the buzzer sound whenever the button is pressed, we must wire them properly to our board. Ensure your hardware is powered down before making these connections.

Step 1: Wiring the Push-button (Input)

- Connect the **GND** (Ground) pin of the push-button to the **GND** pin on your controller board.
- Connect the **VCC / V+** pin of the push-button to the **3.3V / 5V** power pin.
- Connect the **OUT / SIG** (Signal) pin of the push-button to Digital Pin **P1**.

Step 2: Wiring the Buzzer (Output)

- Connect the **GND** pin of the buzzer to an available **GND** pin on the board.
- Connect the **I/O / SIG** (Signal) pin of the buzzer to Digital Pin **P2**.

3. Guided Activity: Building the Doorbell (15 Mins)

Let's create a program that acts exactly like a home doorbell. When the visitor presses the input button, the buzzer sounds a loud alert.

The Logic Rule:

1. Constantly read the state of Digital Pin **P1** (the Button).
2. **IF** the button state is equal to **1** (Pressed), **THEN** write Digital Pin **P2** (the Buzzer) to **1** (Turn On).
3. **ELSE** (Button is not pressed), write Digital Pin **P2** to **0** (Turn Off).

Teacher Checkpoint

Before kids power on their boards, verify that the signal wires for the button and buzzer are not mixed up. Plugs should be firmly secure to prevent intermittent clicking.

4. Review & Challenges (10 Mins)

Review the student's mastery of the hardware components using these interactive questions:

- **Discussion:** What happens to our circuit loop when we release the button? (Answer: The circuit opens, the signal drops to **0**, and the buzzer turns off).
- **Hardware Challenge:** Can you change the behavior so that the buzzer is always making sound *until* you press the button down to silence it?